

1

Introduction to Computer Programming

OBJECTIVES

- Basic hardware and software concepts
- Programming languages

1.1 Introduction

- **Hardware**

- Various devices comprising computer
- Example: keyboard, screen, memory, CD-ROM, etc.

- **Software**

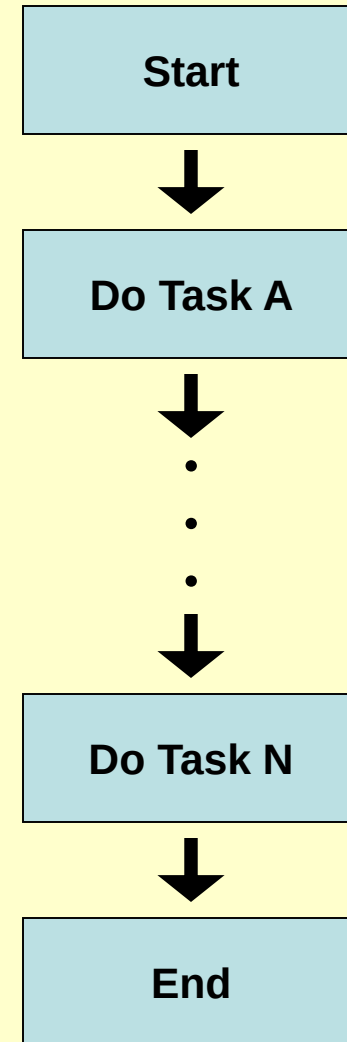
- Instructions to command computers to perform actions and make decisions

- **Programming language**

- A vocabulary and set of grammatical rules for instructing a computer to perform specific tasks

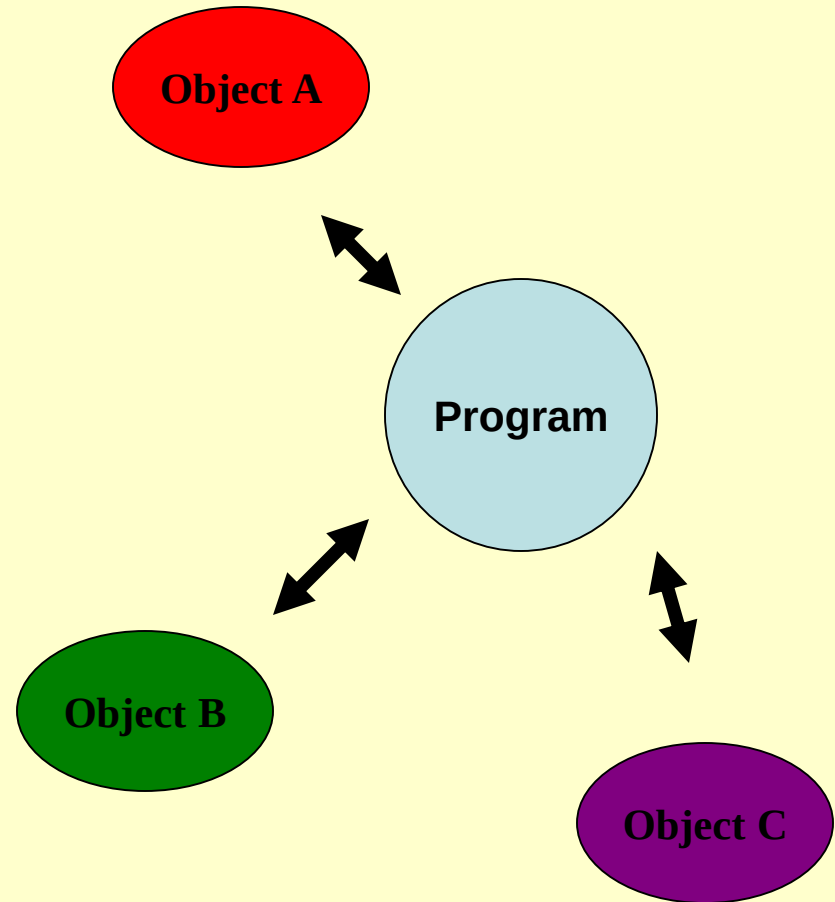
1.1 Introduction

- **Structured programming**



1.1 Introduction

- **Object-oriented programming**



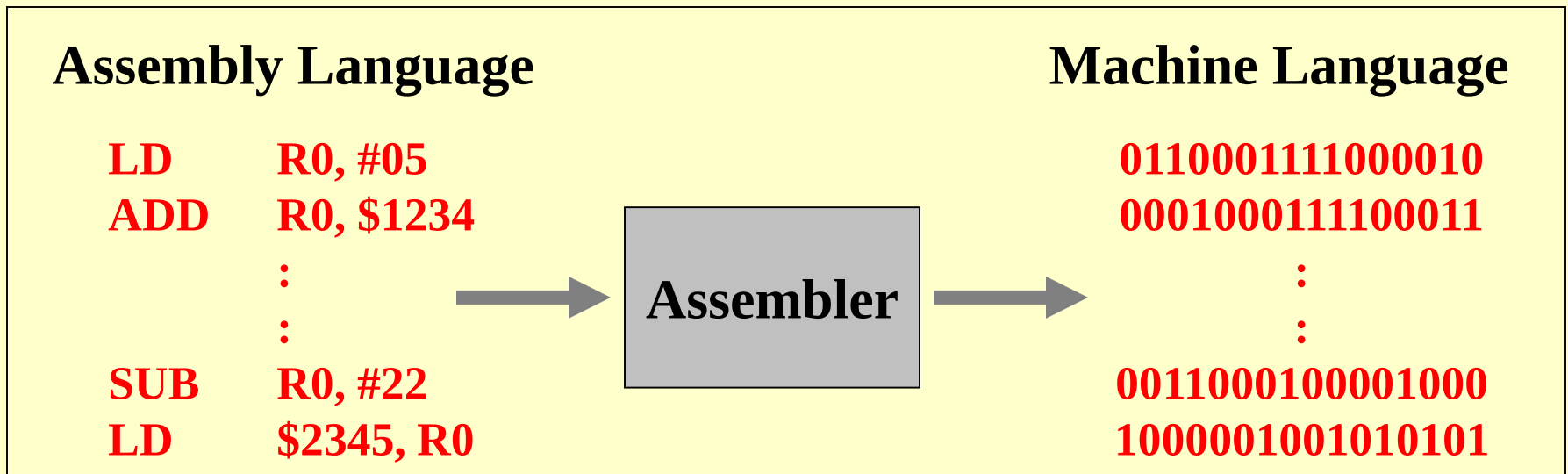
1.7 Three Type of Languages

- **Computer can only understand 0 and 1.**
- **The most direct way to communicate with the computer is to use 0 and 1 $\Rightarrow$ machine language .**
- **Machine language**
 - **“Natural language” of computer**
 - **Defined by hardware design**
 - **Example**
01000101111
11011110001
10011110010

1.7 Three Type of Languages

- **Assembly Language**

- Is created to help human instructs CPU to work.
- By using a tool called **Assembler**, assembly language program can be converted into machine language program.



1.7 Three Type of Languages

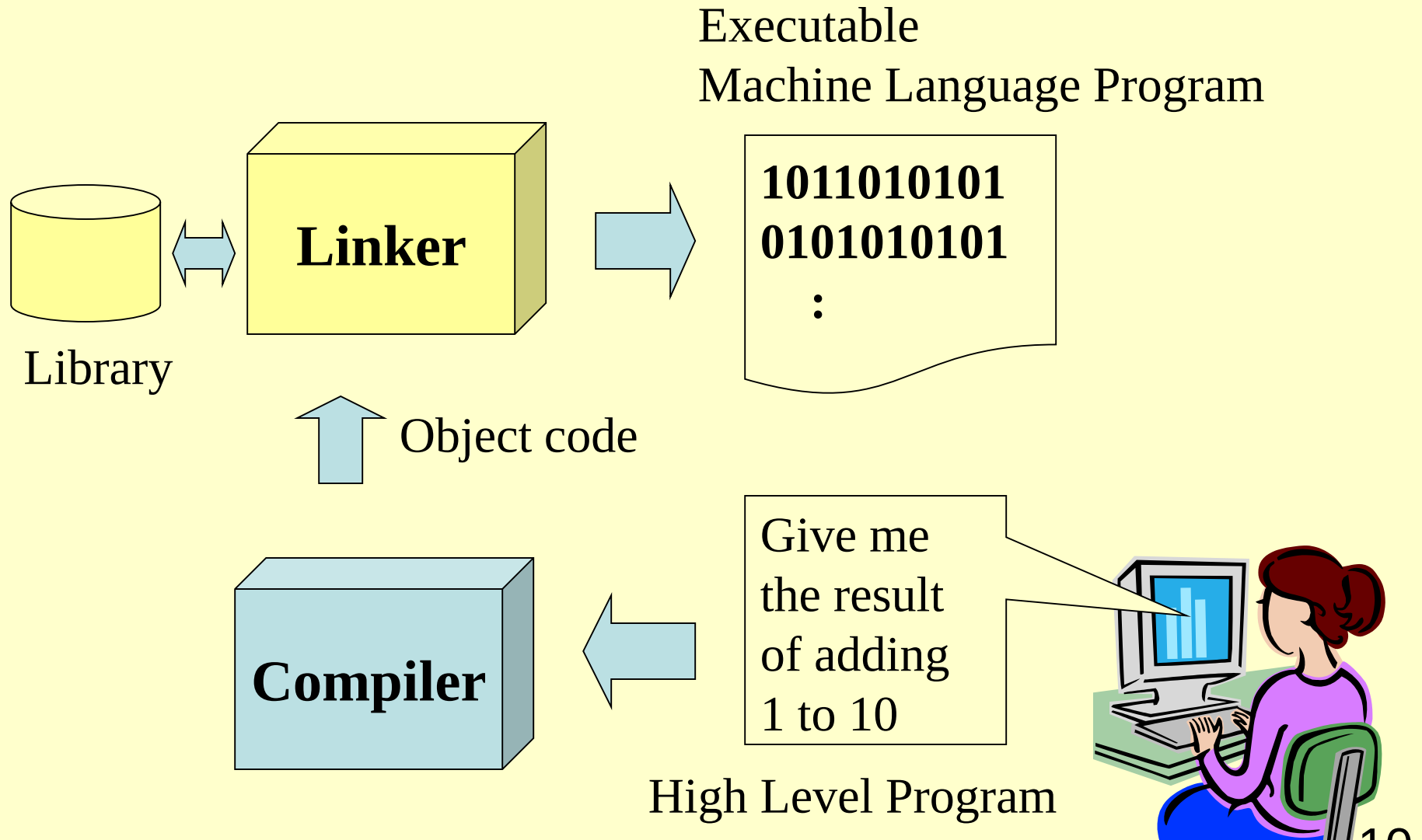
- **Assembly language programming is still too complicated for general users.**
- **A high level language is required to close the gap between human and computers.**

1.7 Three Type of Languages

- **High-level languages**

- Similar to everyday English
- Single statements accomplish substantial tasks.
- Convert to machine language by translator programs (compilers).

1.7 Three Type of Languages



Class Work

- **The level of a computer language most convenient for you to write programs quickly and easily is**

_____ .

- **The only language that a computer directly understands is called that computer's**

_____ .

1.8 History of C and C++

- **History of C**
 - Evolved from BCPL and B
 - Developed by Bell Laboratories
 - Development language of UNIX
 - Hardware independent

1.8 History of C and C++

- **History of C++**
 - **Extension of C**
 - **Also developed by Bell Laboratories**
 - **Provides new features to “upgrade” C**
 - **Provides capabilities for object-oriented programming**

1.10 History of Java

- **Java**
 - **Designed by Sun Microsystems**
 - **Used for:**
 - **Creating Web pages with dynamic content**
 - **Developing large-scale enterprise applications**
 - **Enhancing World Wide Web server functionality**
 - **Providing applications for consumer devices (mobiles, etc.)**

1.11 Other Programming Languages

- **FORTRAN**
 - **FORmula TRANslator**
 - **Used in engineering applications**
- **COBOL**
 - **COmmon Business Oriented Language**
 - **Used in business software**

1.11 Other Programming Languages

- **Pascal**
 - **Designed to teach structured programming**
- **Ada**
 - **Capable of multitasking**

1.12 Other Programming Languages

- **BASIC**
 - **Beginner's All-Purpose Symbolic Instruction Code**
 - **Familiarize novices with programming techniques**
- **.NET platform**
 - **Provides developers with capabilities**

1.12 Other Programming Languages

- **Visual Basic .NET**
 - Based on BASIC
- **Visual C++**
 - Based on C++
- **C#**
 - Based on C++ and Java

1.17 UML

- **Unified Modeling Language (UML)**
 - Graphical language that uses common notation
 - Allows developers to represent object-oriented designs

Until Next Time

- **Review Chapter 1**
- **Preview Chapter 2**